Contents

EEE 554 Lecture #12

Complex Random Vectors

A complex n-vector $Z \in \mathbb{C}^n$ is composed of a real part X and an imaginary part Y, both of which are real n-vectors:

$$Z = X + iY = \begin{bmatrix} x_1 + iy_1 \\ \vdots \\ x_n + iy_n \end{bmatrix}$$

The **Hermitian transpose** $Z^\dagger$ (also denoted as Z^H) is the transpose of the vector with its elements replaced by their complex conjugates:

$$Z^\dagger = [x_1 - iy_1, \dots, x_n - iy_n]$$

The **Euclidean norm** of a complex vector is defined using this Hermitian transpose:

$$||Z|| = (Z^\dagger Z)^{1/2} = \sqrt{\sum_{k=1}^{n} (x_k^2 + y_k^2)} \geq 0$$

Mean and Expectations

The **mean** of a complex random vector is the sum of the expectations of its real and imaginary components:

$$M_Z = E[Z] = E[X] + iE[Y]$$

Expectation remains a **linear operator** for complex vectors: * $E[AZ + B] = AE[Z] + B$ for deterministic matrices A and vectors B * $E[Z_1 + Z_2] = E[Z_1] + E[Z_2]$

Complex Covariance Matrix

The **covariance matrix** of a complex random vector Z is the $n \times n$ matrix defined as:

$$C_Z = E[(Z - M_Z)(Z - M_Z)^\dagger]$$

Expanding this into real and imaginary covariance and cross-covariance matrices yields:

$$C_Z = C_X + C_Y + i(C_{YX} - C_{XY})$$

Note that C_Z is **Hermitian** ($C_Z = C_Z^\dagger$) and **positive semi-definite**.

Proper Complex Random Vectors

A complex random vector is called **Proper** if the following two conditions are met: 1. The real and imaginary parts have equal covariance: $C_X = C_Y$ 2. The cross-covariance matrix is skew-symmetric: $C_{YX} = -C_{XY}$

For proper vectors, the covariance matrix simplifies to:

$$C_Z = 2(C_X + iC_{YX})$$

Furthermore, for a proper vector, the **pseudo-covariance** (or complementary covariance) is zero:

$$E[(Z - M_Z)(Z - M_Z)^T] = 0$$

Multivariate Complex Normal Distribution

A proper complex random vector follows a **Multivariate Complex Normal (Gaussian) Distribution** if its probability density function (PDF) is:

$$f_Z(z) = \frac{1}{\pi^n \det(C_Z)} \exp\left\{-(z - M_Z)^\dagger C_Z^{-1}(z - M_Z)\right\}$$

This distribution is often denoted as $Z \sim \mathcal{CN}[M_Z, C_Z]$.